INVERTED MAML RESEARCH TEAM — MULTI-AGENT ARCHITECTURE

Autonomous Multi-Agent Organization, Responsibilities, and Adversarial Scientific Protocol

Patent Pending: SE 2630397-4 | **Hardware:** Trion T120 SoC + T20 Coprocessor | **Date:** 2026-10-13 18:58:26 CEST

The Inverted MAML research framework operates as a specialized multi-agent hierarchy designed to decouple strategic direction, physical hardware execution, mathematical modeling, and rigorous adversarial validation. This structure guarantees reproducible research, prevents confirmation bias, strictly enforces hardware safety (heaters disarmed, storage budgets < 100 KB), and ensures that physical jumper configurations (Modes A, B, and C) are respected at all times.

1. Multi-Agent Functional Block Architecture

PILLAR 1: STRATEGIC LEADERSHIP	PILLAR 2: HARDWARE & EMULATION	PILLAR 3: ALGORITHM & DATA	PILLAR 4: ADVERSARIAL AUDIT
maml_team_leader Overall picture, roadmap orchestration, and datacenter perspective. **maml_public_rep** Public liaison, human reporting, and PDF dossier generation. **maml_research_planner** Firmware changelog maintainer, scheduling, and gatekeeper.	**maml_hardware_expert** *Primary Custodian for Jumper Modes A/B/C.* Enforces no-training rule in Mode B. Manages heater lockout (HEAT=0). **maml_hardware_emulator_expert** Models latent physical mismatch (W_hidden), temporal capacitor droop, and 2-quadrant warp.	**maml_nn_specialist** Inverted MAML outer loop 40, inner loop 41, and Hessian dot-product curvature. **maml_data_crawler** UDP binary telemetry (0x4D414D4C), automated scans, disk budget guard. **maml_data_analyzer** Computes ENOB, MSE loss trends, linearity slopes, and spans.	**maml_bug_methodology_critic** *'The Devil's Advocate'*: Continuously challenges ENOB metrics, DC offsets, and measurement validity. **maml_scientific_proof_expert** Formal mathematical verification, Null Hypothesis testing ($p < 1e-15$, $d = 20.4$), and peer-review proof dossiers.

2. Operational Responsibilities & Inter-Agent Protocols

Agent Name	Primary Mandate	Key Deliverables & Enforced Constraints
maml_team_leader	Synthesizes overall research trajectory, balances model architecture with physical silicon limits.	Ensures datacenter focus (TOPS/W, low latency, 94% duty cycle inference). Sets campaign targets.
maml_hardware_expert	Hardware Custodian: Jumper Modes (A/B/C), analog multi-carrier buses, and thermal safety.	**Enforces Jumper Mode B strict no-training rule.** Interlocks HEAT_1..4_EN to 0% duty at all times.
maml_hardware_emulator_expert	Builds high-fidelity software surrogates of analog non-idealities.	Models W_hidden mismatch (σ=59 LSB), pedestal bias (+48 LSB), and 10 ms capacitor droop (analog_hardware_surrogate.py).
maml_nn_specialist	Oversees non-backpropagation Hadamard gradient descent (Eqs. 1–7).	Monitors Hessian dot-product alignment (+16,380 saturation in Q1.14) and basin spread (≤ 5 units).
maml_bug_methodology_critic	Adversarial Skeptic: Audits telemetry and methodology for false positives.	Scrutinizes ENOB normalization, ADC settling latencies, and cross-talk artifacts (MEASUREMENT_METHODOLOGY_CRITIQUE.md).
maml_scientific_proof_expert	Formulates mathematical proofs and statistical verification dossiers.	Performs paired t-tests, rejects H_0 ($p < 10^{-15}$, Cohen's $d = 20.36$), and authors formal dossiers (SCIENTIFIC_PROOF_DOSSIER.md).
maml_research_planner	Maintains system changelog, software builds, and execution sequence.	Authors firmware_changelog.md with exact release dates, bitstream hashes, and verification steps.

3. Core Operational Directives

- **1. Zero Unsupervised Thermal Risk:** Power heaters (HEAT_1..4_EN) remain permanently disarmed (duty = 0) to protect hardware.
- **2. Strict Disk Conservation:** All telemetry crawls and benchmark logs use atomic rotating buffers strictly capped under 100 KB.
- **3. Hardware-Informed Adaptation:** Inverted MAML adapts purely through forward Hadamard projections—no digital backpropagation.
- **4. Jumper Discipline:** Mode B (bare silicon) is purely diagnostic; compensation weights are frozen to prevent divergence.